Nanotechnology

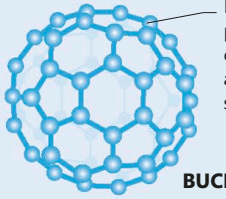
Nanotechnology is the area of technology that involves creating and manipulating matter and objects at an extremely small scale, known as the nanoscale.

The nanoscale

Nanoscale objects measure between 1 and 100 nanometers (nm), where 1 nm is a billionth of a meter. Some molecules, such as glucose, antibodies (large protein molecules), and viruses are nanoscale objects.

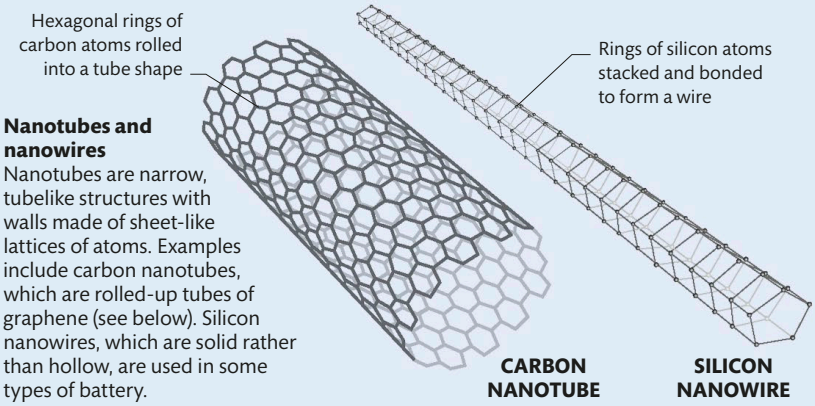
Nanomaterials

A nanomaterial is any material or object with at least one dimension (length, width, or height) smaller than 100 nm. Some nanomaterials occur naturally—such as smoke particles, spider silk, and some butterfly wing scales—while others are deliberately created with unique properties. For example, gold nanoparticles can be engineered so that they emit a burst of heat when illuminated with light, a property that can be used to destroy cancer cells.



Nanoparticles

A nanoparticle is an object with all three dimensions on the nanoscale. Many nanoparticles have unusual properties due to their size or shape; for example, the hollow structure of buckyballs means they could carry other molecules inside.

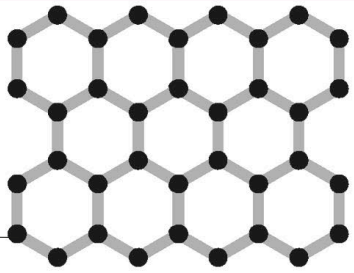


Nanotubes and nanowires

Nanotubes are narrow, tubelike structures with walls made of sheet-like lattices of atoms. Examples include carbon nanotubes, which are rolled-up tubes of graphene (see below). Silicon nanowires, which are solid rather than hollow, are used in some types of battery.

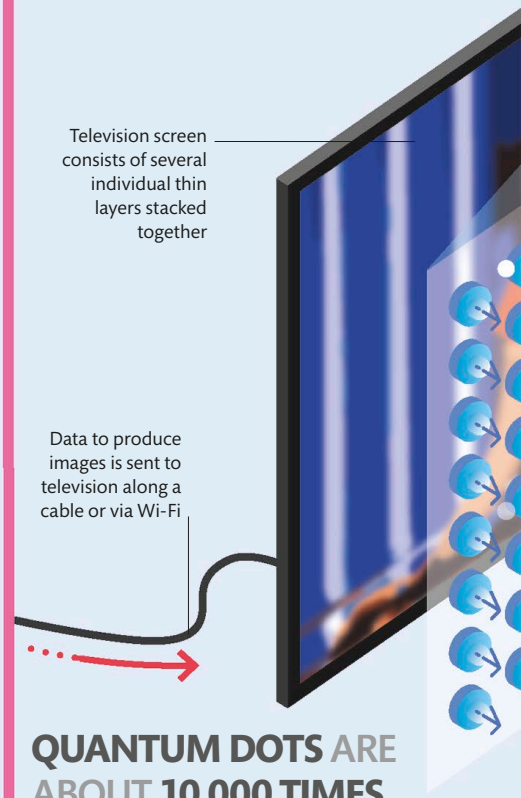
GRAPHENE

Graphene is a one-atom-thick layer of carbon atoms, arranged in a hexagonal (honeycomb) lattice. It is very stiff in all directions and is the strongest material ever tested. Graphene is also an excellent conductor of heat and electricity.



Quantum dot television

Some television screens use nanoparticles in the form of quantum dots to achieve a brighter, sharper, more colorful image. In these displays, an array of quantum dots is positioned on top of the LED and liquid crystal layers. When the differently sized dots are stimulated with blue light from the LEDs, they emit pure red and green light. The combination of red, green, and blue light from each pixel of the screen is perceived as a single color.



QUANTUM DOTS ARE ABOUT 10,000 TIMES NARROWER THAN A HUMAN HAIR

